Class5. Data Viz with ggplot

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Background

There are lots of ways to make plots in R. These include so-called "base R" (like the `plot()` and add on packages like **ggplot**)

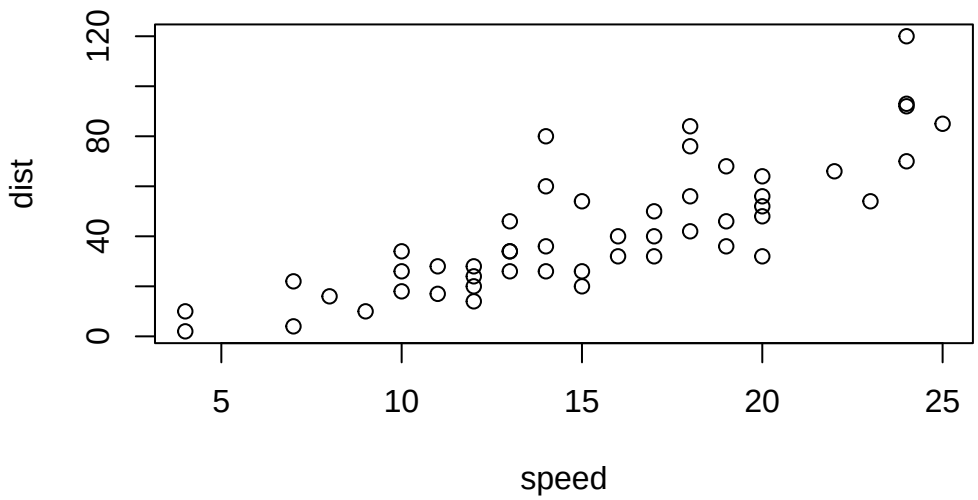
Let's make the same plot with these two graphics system. We can use the inbuilt `cars` dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

With "base R" we can simply:

```
plot(cars)
```



Now let's try ggplot, First we need to stall the package using the right code `install.packages("ggplot2")`.

> **N.B.** We never run the `install.packages()` in the code chunck otherwise we need to re-install every time we render our document.

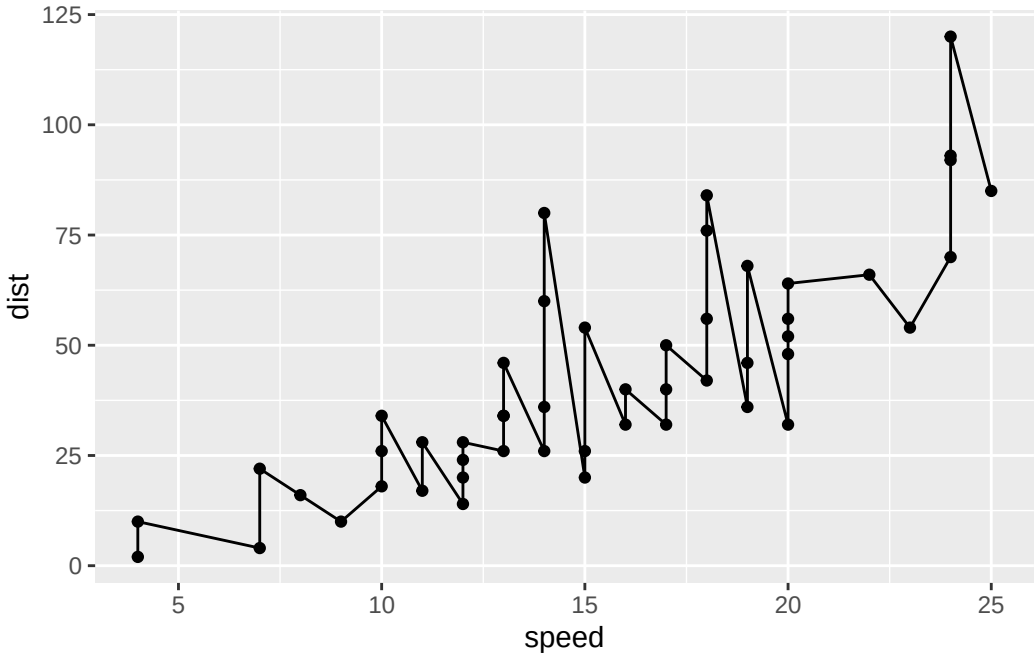
Every time we want to use an add-on package, we need to load it up with a call to `library()`

```
library(ggplot2)
ggplot(cars)
```

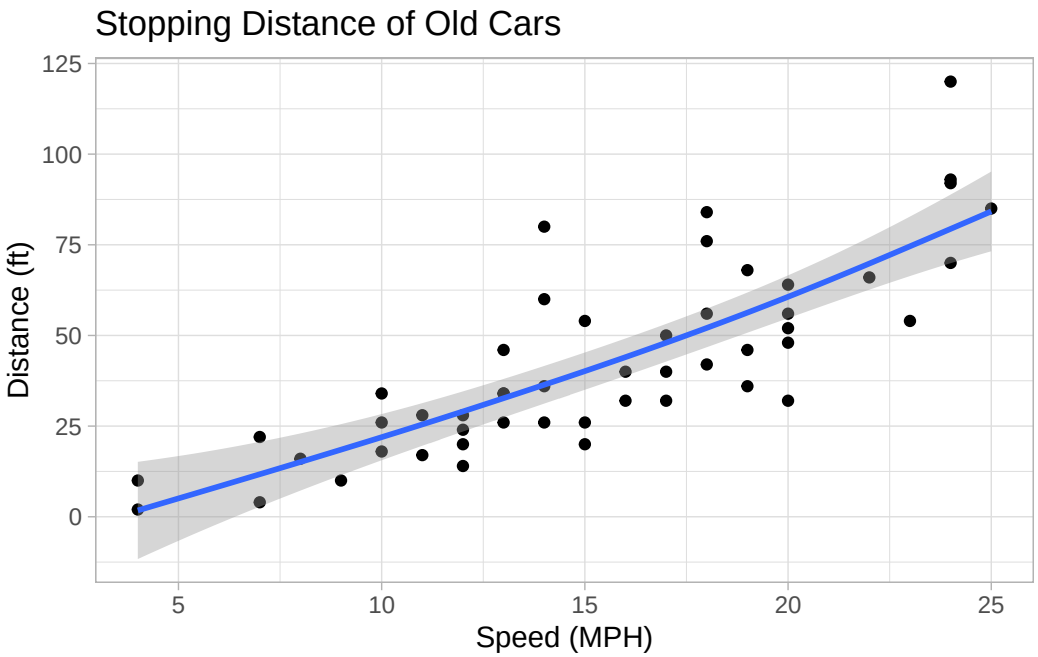
Every ggplot needs at least 3 things:

1. The **data** i.e. stuff to plot as a data.frame
2. The **aes** for aesthetics that map the data to the plot
3. The **geom_** or geometry i.e. the plot type such as points, lines etc.

```
ggplot(cars) +
  aes(x=speed, y = dist) +
  geom_point() +
  geom_line()
```




```
ggplot(cars) +
  aes(x=speed, y = dist) +
  geom_point() +
  geom_smooth(method = "gam", formula = y ~ s(x)) +
  labs(x="Speed (MPH)",
       y="Distance (ft)",
       title = "Stopping Distance of Old Cars") +
  theme_light()
```



Gene Expression Plot

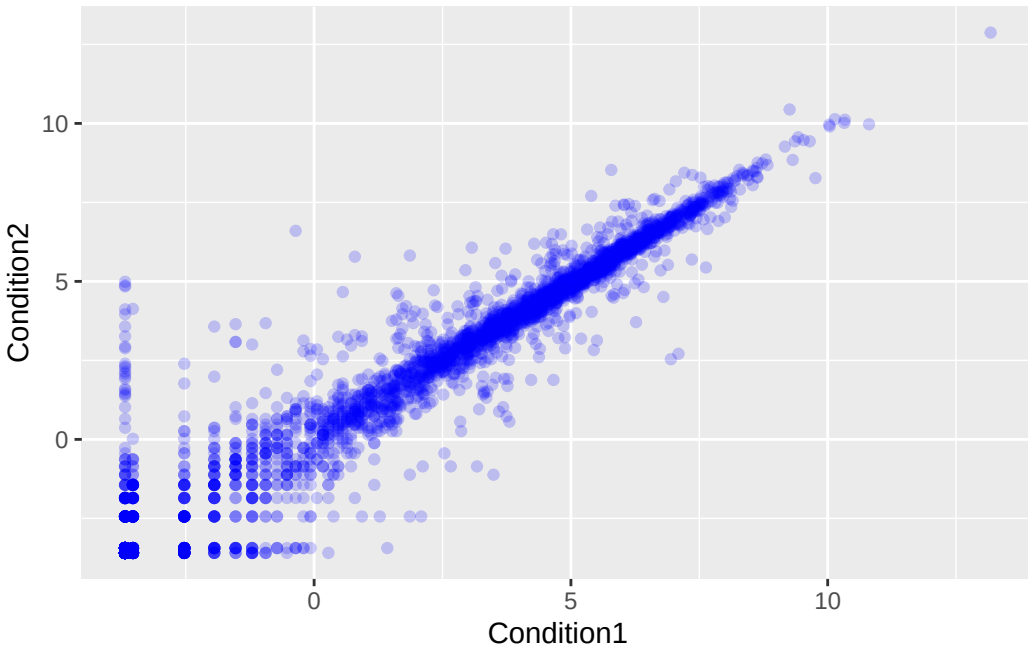
Read some data on the effectes of GLP-1 inhibitor (drug) on gene expression values:

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

Version 1 plot - start simple by getting some ink on page.

```
ggplot(genes)+
  aes(Condition1,Condition2)+
  geom_point(col="blue", alpha = 0.2)
```

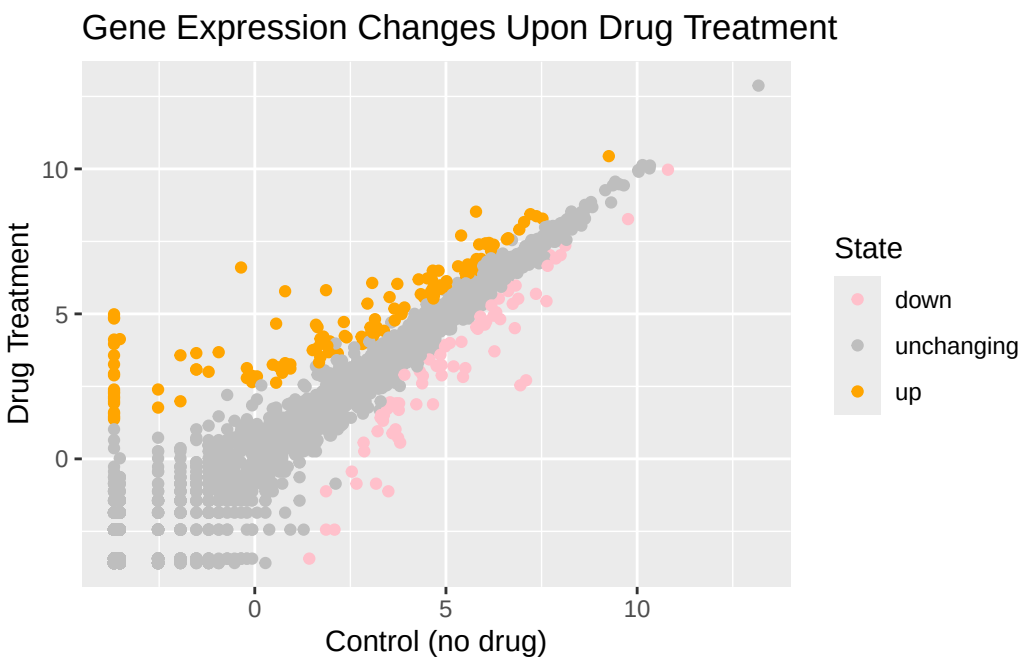


Let's color by `State` up, down or no change.

```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

```
ggplot(genes)+
  aes(x=Condition1,y=Condition2,col = State)+
  geom_point()+
  scale_color_manual(values = c("pink","gray","orange"))+
  labs(x = "Control (no drug)", y="Drug Treatment", title = "Gene Expression Changes Upon Dru
```



Going futher with gapminder

Here we explore the famous `gapminder` dataset with some custom plots.

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Question. How many rows does this dataset have?

```r
nrow(gapminder)
```

```
[1] 1704
```

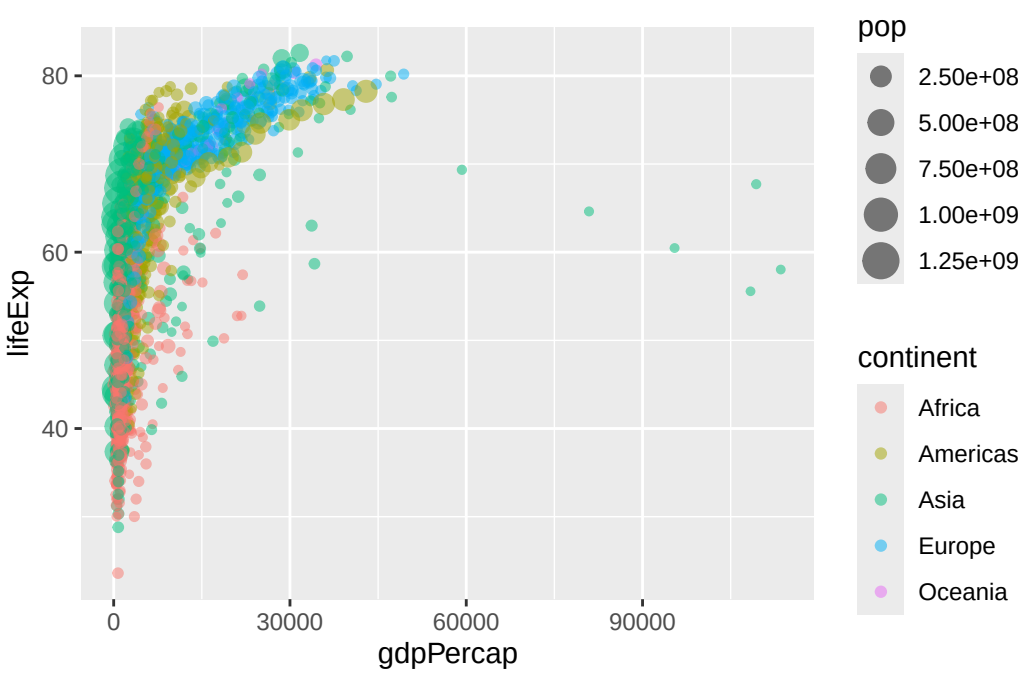
Question. How many different continents are in this dataset?

```r
table(gapminder$continent)
```

```
  Africa Americas     Asia   Europe  Oceania 
     624      300      396      360       24 
```

Version 1 plot gdpPercap vs LifeExp for all rows

```r
ggplot(gapminder)+
  aes(x = gdpPercap, y = lifeExp, col = continent, size = pop)+
  geom_point(alpha=0.5)
```



Version 2: color the points by value of population:

```r
ggplot(gapminder) +
  aes(x = gdpPercap, y = lifeExp, color = pop) +
  geom_point(alpha=0.5) +
  scale_size_area(max_size = 10)
```

Q. produce the plot for 1957 vs 2007

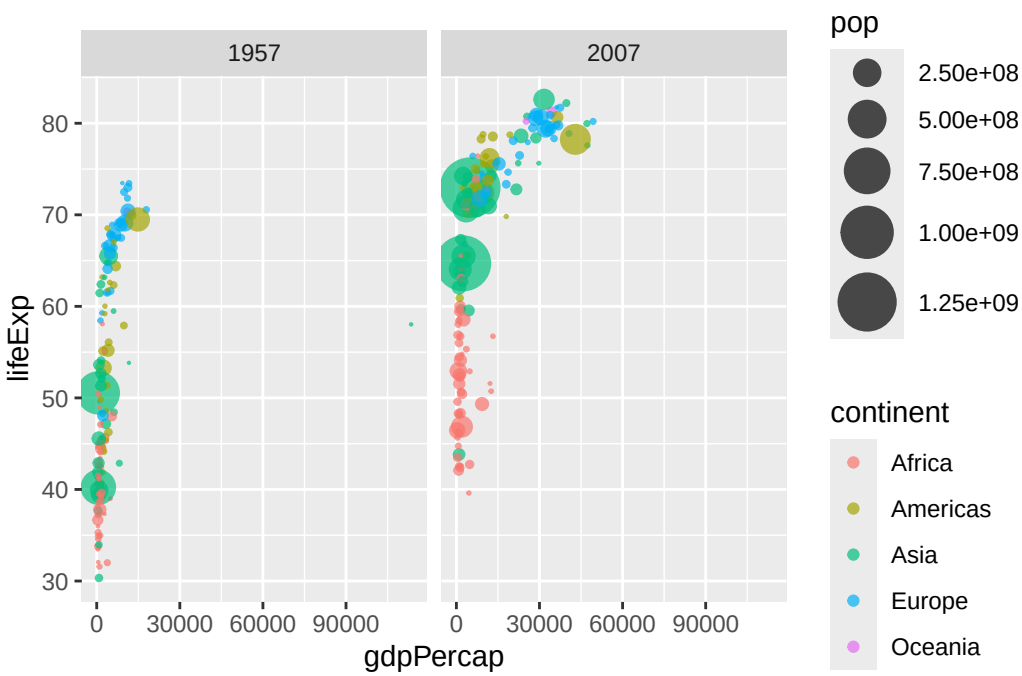
```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

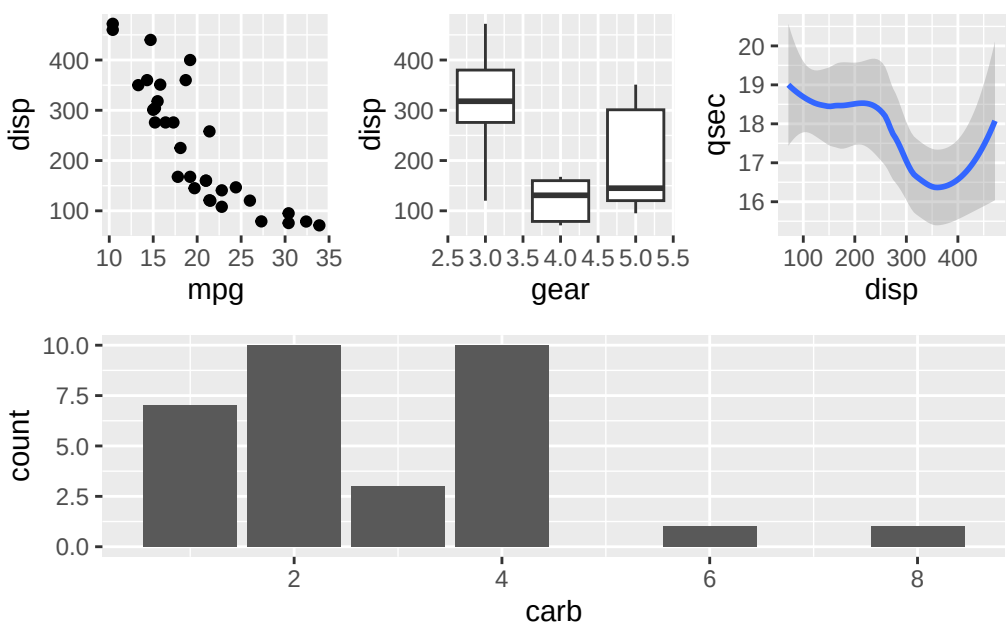
```r
library(gapminder)
```

```
Attaching package: 'gapminder'

The following object is masked _by_ '.GlobalEnv':

    gapminder
```

```r
gapminder_1957 <- gapminder %>%
  filter(year %in% c(1957, 2007))

ggplot(gapminder_1957) +
  geom_point(aes(x = gdpPercap, y = lifeExp, color=continent,
                 size = pop), alpha=0.7) +
  scale_size_area(max_size = 10) +
  facet_wrap(~year)
```



##Combining plots

```r
library(patchwork)

# Setup some example plots
p1 <- ggplot(mtcars) + geom_point(aes(mpg, disp))
p2 <- ggplot(mtcars) + geom_boxplot(aes(gear, disp, group = gear))
p3 <- ggplot(mtcars) + geom_smooth(aes(disp, qsec))
p4 <- ggplot(mtcars) + geom_bar(aes(carb))

# Use patchwork to combine them here:
(p1 | p2 | p3) /
      p4
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```